BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Final | Semester: Fall 2023
Duration: 2 hours | Full Marks: 50

CSE 320/EEE361/ECE361: Data Communications

Answer the following questions.
Figures in the right margin indicate marks.

SET B

Name:	ID:	Section:

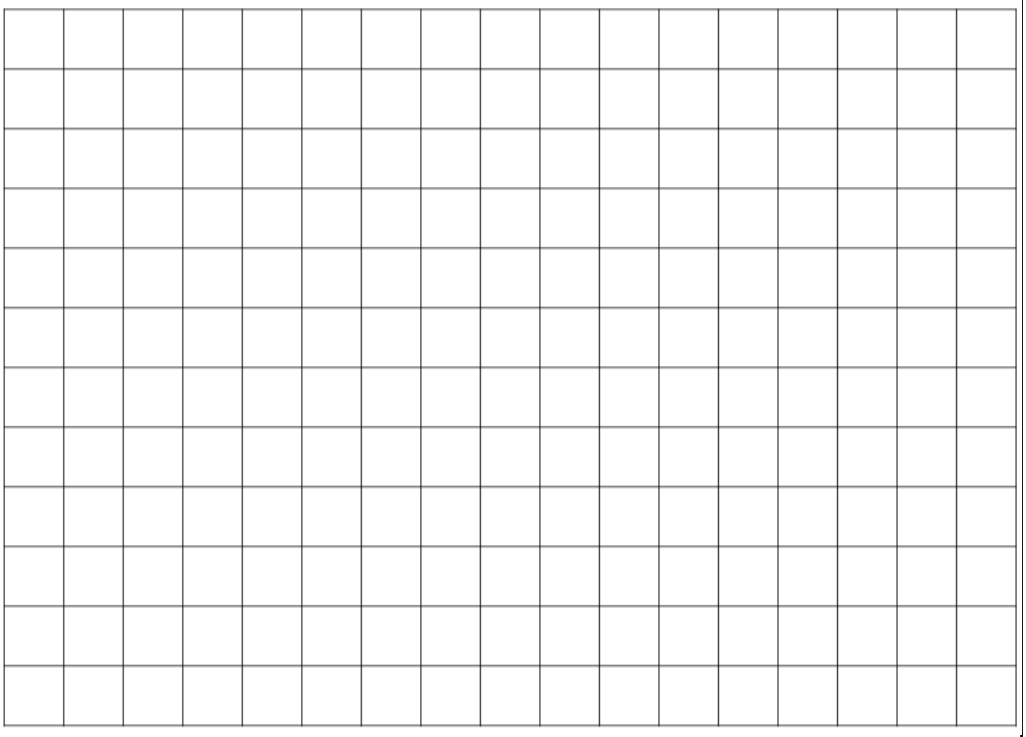
1. [CO3]

a) **Consider** there are five channels mentioned in the below table which are to be multiplexed in a way that the number of channels will **decrease**. If **one** synchronization bit is added to each frame and the interleaved unit from each channel is **2 bits**, **write** down the answers of the following questions. [2+8]

[***Hint: Convert the source speed to a common unit like Kbps (1000bits) or bps first and show the calculation for each question below***]

Channel Name	Data rate
A	20000 Character/s
B	20 KBytes/s
C	160 Kbps
D	10000 Character/s
E	10 KBytes/s

i. **Diagrammatically** represent the physical multiplexer with necessary details.
ii. In what kind of TDM, the channels are reduced?
iii. What is the **input bit** duration in **seconds**?
iv. What is the size of a frame in **bits**?
v. What is the frame rate **(FPS)**?
vi. What is the duration of a **frame** in **seconds**?
vii. What is the output data rate or transmission rate of the medium in **bps**?
viii. What is the output slot duration in **seconds**?
ix. How many overhead **bits** are passing through the medium per second? *[hint: frame rate]*

1. **[CO3]**	b)	**Sketch** the **Original Signal, Spread Code,** and **Spread Signal** from the following Original Signal (**101**) and the given spreading code (**01010**) using the concept of DSSS. You can use the **NRZ line coding scheme** (zero means low and one means high voltage) to draw the signals. 	[3]
[CO3]	c)	What is the **drawback** of using DSSS?	[2]
2. **[CO4]**	a)	**Compare** between Coaxial cable and Optical fiber cable types (mention two points).	[2]
[CO4]	b)	**Identify** the transmission medium for the scenario given below: i. Commonly used for telephone lines and low-speed network connections. ii. RJ-45 connector is used. iii. It has a central core conductor of solid wire enclosed in an insulating sheath.	[3]
3. **[CO5]**	a)	Assume a packet is made only of four **8-bit** words $(124)_{10}$, $(20)_{10}$, $(221)_{10}$ **and** $(100)_{10}$. i. **Show** the checksum at the sender. ii. If the **second data** item is changed to $(45)_{10}$ and the **third data** item is changed to $(196)_{10}$ during transmission, check if any error can be detected at the **receiver site**. **Explain** your findings.	[5]

<table>
<tr><td>3.
[CO5]</td><td>b)</td><td>Suppose two channels are using the following block coding scheme for error detection and correction.
i. How many bits can it successfully detect and correct errors using this scheme?
ii. The receiver received the codeword 11110010 that contains 1 bit error. Determine if the original dataword can be retrieved from it or not?

<table><tr><th>Dataword</th><th>Codeword</th></tr><tr><td>00</td><td>01110111</td></tr><tr><td>01</td><td>11110000</td></tr><tr><td>10</td><td>10111010</td></tr><tr><td>11</td><td>00001101</td></tr></table></td><td>[5]</td></tr>
<tr><td>[CO5]</td><td>c)</td><td>Assume the following cases for a sender and a receiver device that use CRC for error control
▪ Dataword: $x^6 + x^5 + x^3 + x^2$
▪ Divisor: $x^4 + x^2 + 1$

Show the sent codeword using CRC polynomial division.</td><td>[5]</td></tr>
<tr><td>4.
[CO5]</td><td>a)</td><td>In CSMA/CA, what is done by a station if the channel is found idle? Write down the purpose of doing so.</td><td>[2]</td></tr>
<tr><td>[CO5]</td><td>b)</td><td>Identify the name of media access protocols for the scenario given below:
i. Stations transmit at any time and check for collisions.
ii. Stations can transmit only at the beginning of the time slot.
iii. Each station sends a frame whenever it has a frame to send.
iv. Stations transmit using different frequencies simultaneously.
v. Stations transmit data after receiving permission from a central controller.</td><td>[5]</td></tr>
<tr><td>[CO5]</td><td>c)</td><td>For the 2nd attempt (k=2) in an Ethernet-based MAC protocol, what are the possible Back-off times(T_B) if the Transmission time (T_{fr}) is 3 ms?</td><td>[3]</td></tr>
<tr><td>[CO5]</td><td>d)</td><td>In a CSMA/CD network running with a bandwidth of 2 Gbps over a cable, the signal speed in the cable is 40000 km/sec and the minimum frame size is 6000 bits. Calculate the distance covered by the signal.</td><td>[5]</td></tr>
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